

(Unit-3)

Introduction to web analytics

(Data collection, Weblogs & Page tagging, Key metrics . ■ Unique visitors, Pageviews, Pages/visit, Avg Visit Duration, Bounce rate, New Visits. Give me with examples)

Semrush Tool| Website Auditing| Organic Traffic| Paid Traffic

https://www.youtube.com/watch?v=z-5jhpia45g&list=PL5hA7O8RI2bMH9S_-KbY7K-73hlrAyTHF&index=46

Light House Tool| How to use Light House Tool for Webpage Auditing| Webpage performance

https://www.youtube.com/watch?v=PaaFYkIJjCU&list=PL5hA7O8RI2bMH9S_-KbY7K-73hlrAyTHF&index=47

Website Analytics| Social Media and Webanalytics in hindi

https://www.youtube.com/watch?v=VwwEjN6toBY&list=PL5hA7O8RI2bMH9S_-KbY7K-73hlrAyTHF&index=54

What is GoogleTag Manager| Google Analytics| How to Use Google Analytics|Website performance

https://www.youtube.com/watch?v=bcX1mrPHoII&list=PL5hA7O8RI2bMH9S_-KbY7K-73hlrAyTHF&index=55
https://www.youtube.com/watch?v=A546lwOg_4s

Google Analytics Live Demo

https://www.youtube.com/watch?v=PRudjs-vsrg&list=PL5hA7O8RI2bMH9S_-KbY7K-73hlrAyTHF&index=56

A/B and Multivariate Testing in Digital Marketing

https://www.youtube.com/watch?v=cho1WzBOCoE&list=PL5hA7O8RI2bMH9S_-KbY7K-73hlrAyTHF&index=57

Attribution Model| Single-Touch| First Touch| Last Touch| Multi-Touch| Linear|Time-Decay

https://www.youtube.com/watch?v=ucroCC7TLdU&list=PL5hA7O8RI2bMH9S_-KbY7K-73hlrAyTHF&index=58

Google Tag Manager

<https://www.youtube.com/watch?v=8MGOJAiZWa8>

1. Data Collection

This is how you gather information about visitors to your website. Two common methods are **Weblogs** and **Page Tagging**.

a) Weblogs

A Weblog (or Server Log File) is a file automatically created by the web server whenever someone requests a webpage or resource.

- **What it is:** Raw log files maintained by the web server that store every request made (pages, images, downloads).
- **How it works:** The server records the visitor's IP address, date/time, requested URL, browser type, etc.
- **Example:**

If you run www.chahanstore.com, your server log might have an entry like:

```
192.168.1.15 - - [12/Aug/2025:11:25:45] "GET /shoes.html HTTP/1.1" 200 5320
```

This tells you someone visited shoes.html at that time.

Example

A visitor opens:

```
https://www.xyzstore.com/shoes
```

The server records:

```
IP Address: 192.168.1.25
Date: 07-Aug-2026 10:30 AM
Page: /shoes
Method: GET
Browser: Chrome
Operating System: Windows 11
Status Code: 200
```

What Information is Stored?

- Visitor IP Address
- Date & Time
- Requested URL
- HTTP Method (GET/POST)
- Browser Type

- *Operating System*
- *Status Code (200, 404, 500)*
- *Bytes Sent*
- *Referring Website*

Advantages

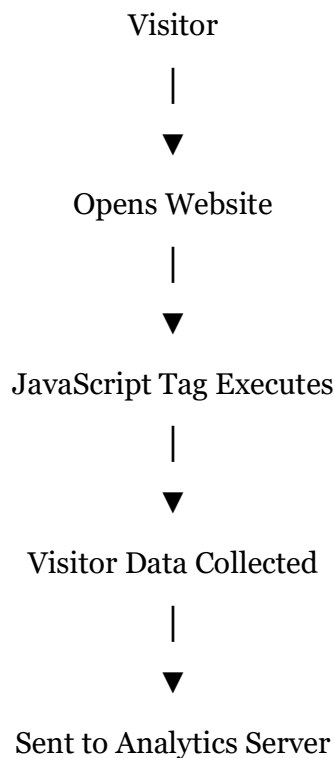
- *No additional code required.*
- *Tracks every server request.*
- *Useful for SEO and crawl analysis.*
- *Works even if JavaScript is disabled.*
- *Good for security monitoring.*

Disadvantages

- *Cannot accurately track user actions like button clicks or scrolling.*
 - *Difficult to identify unique users.*
 - *Requires technical knowledge to analyze.*
-

b) Page Tagging

Page Tagging is a tracking method where a small JavaScript code is placed on every webpage. When a visitor opens the page, the script sends data to an analytics platform.





Reports & Dashboard Generated

- **What it is:** Adding small pieces of JavaScript code (tracking tags) to your webpages to collect visitor data in real time.
- **How it works:** When a visitor loads the page, the tag sends data to an analytics platform (e.g., Google Analytics).
- **Example:**

Example

A website contains:

HTML

```
<script>  
Google Analytics Tracking Code  
</script>
```

When a visitor opens:

<https://www.xyzstore.com/shoes>

The tag sends information like:

- Page Viewed
- Visitor Location
- Device Type
- Browser
- Session Duration
- Clicks
- Purchases
- Bounce Rate

Example

You add Google Analytics tracking code to your site:

html

```
<script async src="https://www.googletagmanager.com/gtag/js?id=UA-123456"></script>
```

Now you can see traffic in Google Analytics instantly.

Advantages

- Easy to measure marketing campaigns.
- Provides real-time reports.
- Tracks user interactions such as clicks, downloads, and form submissions.
- Measures conversions and sales.
- Easy integration with advertising platforms.

Disadvantages

- Requires JavaScript to be enabled.
- Ad blockers or privacy settings may block tracking.
- Requires tracking code on every page.

Weblogs & Page Tagging in Digital Marketing

Both **Weblogs** and **Page Tagging** are methods used to collect website visitor data. They help marketers understand how users interact with a website.

Feature	Weblogs (Log File Analysis)	Page Tagging
Definition	A method that records all website requests in the web server's log files.	A method that uses JavaScript tags embedded in web pages to collect visitor data.
Where Data is Stored	Web server log files.	Analytics platform (e.g., Google Analytics).
How it Works	Every time a visitor requests a webpage, the server automatically records information in a log file.	When a webpage loads, a JavaScript tracking code sends visitor information to the analytics server.
Data Collected	IP address, date/time, browser, operating system, requested page, status code, bytes transferred.	User behavior, page views, clicks, scrolls, events, conversions, device information, location, etc.
Requires JavaScript?	No	Yes
Tracks Bots/Crawlers?	Yes	Mostly No (bots usually don't execute JavaScript).
Real-time Reporting	Limited	Yes
Accuracy	Less accurate for user behavior	More accurate for marketing analysis
Popular Tools	Apache Logs, IIS Logs, Nginx Logs	Google Analytics, Adobe Analytics, Matomo
Best Used For	Server monitoring, security, SEO crawl analysis	Marketing campaigns, user behavior analysis, conversion tracking

1. Data Collection Methods

Method	Definition	How it Works	Example
Weblogs	Raw server log files that store all requests to the website.	Recorded automatically by the web server for each hit.	Log entry: 192.168.1.15 - - [12/Aug/2025:11:25:45] "GET /shoes.html HTTP/1.1" 200 5320
Page Tagging	Adding tracking code (JavaScript) to web pages to collect visitor data.	Code sends user actions to analytics software (e.g., Google Analytics).	<pre><script async src="https://www.googletagmanager.co m/gtag/js?id=UA-123456"></script></pre>

2. Key Metrics (with examples)

a) Unique Visitors

- **Definition:** Number of distinct individuals visiting your site in a given period (counted by unique browser cookies or IPs).
- **Example:**
If the same person visits www.chahanstore.com 5 times in a week from the same device, they are counted as **1 unique visitor**.

b) Pageviews

- **Definition:** Total number of times a page is loaded (including repeats by the same visitor).
- **Example:**
If one visitor views the "Home" page 3 times and the "Shoes" page twice, that's **5 pageviews**.

c) Pages/Visit (Page Depth)

- **Definition:** Average number of pages viewed per visit/session.
- **Example:**
If in 3 visits:
 - Visit 1: 3 pages viewed
 - Visit 2: 2 pages viewed
 - Visit3: 5 pages viewed

$$\text{Pages/Visit} = (3+2+5) / 3 = 3.33 \text{ pages/visit}$$

d) Avg. Visit Duration

- **Definition:** Average time a visitor spends on the site per visit.
- **Example:**
If three visits lasted 2 min, 4 min, and 6 min:
Average = (2+4+6) / 3 = 4 minutes

e) Bounce Rate

- **Definition:** Percentage of single-page visits (visitor leaves without interacting further).
- **Example:**
If 100 people land on your site and 60 leave without clicking another page,
Bounce rate = 60%.

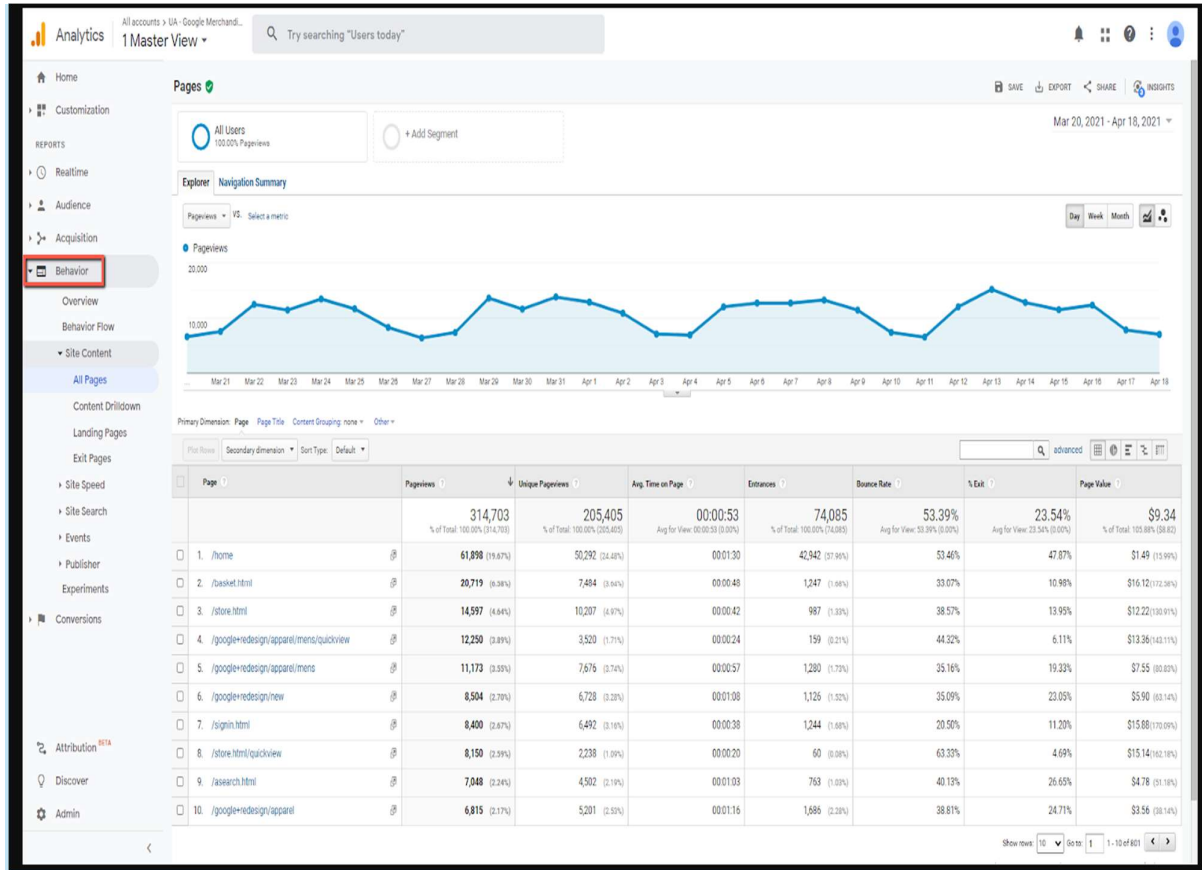
f) New Visits

- **Definition:** Percentage of visitors coming to the site for the first time.
- **Example:**
If you had 200 total visits and 80 came from first-time visitors,
New Visits = $(80/200) \times 100 = 40\%$.

2. Key Metrics in Web Analytics			
Metric	Definition	Formula	Example
Unique Visitors	Number of distinct individuals visiting during a period.	Count of unique browser cookies/IPs.	1 person visits 5 times → counted as 1 unique visitor.
Pageviews	Total number of times a page is loaded.	Count of all page loads.	1 visitor views Home(3x) + Shoes(2x) → 5 pageviews.
Pages/Visit	Avg. number of pages viewed per session.	Total Pageviews ÷ Total Visits.	(3 + 2 + 5) pages in 3 visits → 3.33 pages/visit.
Avg. Visit Duration	Average time spent on site per visit.	Total Time Spent ÷ Total Visits.	(2 + 4 + 6) mins ÷ 3 visits → 4 mins.
Bounce Rate	% of visitors who leave after viewing only 1 page.	(Single-page visits ÷ Total Visits) × 100.	60 out of 100 visitors leave after 1 page → 60% bounce rate.
New Visits	% of first-time visitors.	(First-time Visits ÷ Total Visits) × 100.	80 new visits out of 200 total → 40% new visits.

Types of web analytics Tools

Category	Purpose / Use Case	Examples
Content Analytics Tools	Measure performance of blogs, videos, and social content (shares, engagement, reach).	BuzzSumo, Parse.ly, SEMrush Content Analyzer
Customer Analytics Tools	Understand customer segments, lifetime value, churn, and buying behavior.	Mixpanel, Kissmetrics, Salesforce Customer 360
Usability (UX) Analytics Tools	Track user interactions (clicks, scrolls, heatmaps) and identify usability issues.	Hotjar, Crazy Egg, FullStory
A/B & Multivariate Testing Tools	Test different versions of pages, apps, or features to optimize conversions.	Optimizely, VWO, Google Optimize
SEO Analytics Tools	Improve search ranking, track keywords, backlinks, and website visibility.	Ahrefs, Moz, SEMrush, Google Search Console
General Enterprise Analytics Tools	Business intelligence, dashboards, overall data insights for decision-making.	Google Analytics 4, Adobe Analytics, Tableau, Power BI
Open-Source Web Analytics Tools	Free/community-driven tools for tracking website traffic & behavior.	Matomo (Piwik), Open Web Analytics, Countly
Product Analytics Tools	Analyze in-app behavior, feature usage, product adoption & growth.	Amplitude, Heap, Pendo, Mixpanel



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Vpn login Current Affairs State Bank of India ... 200 Best Current Aff... Welcome to Central... Indian Economy Qui...

Primary Dimension: Page Page Title Other

Plot Rows Secondary dimension Sort Type: Default

advanced

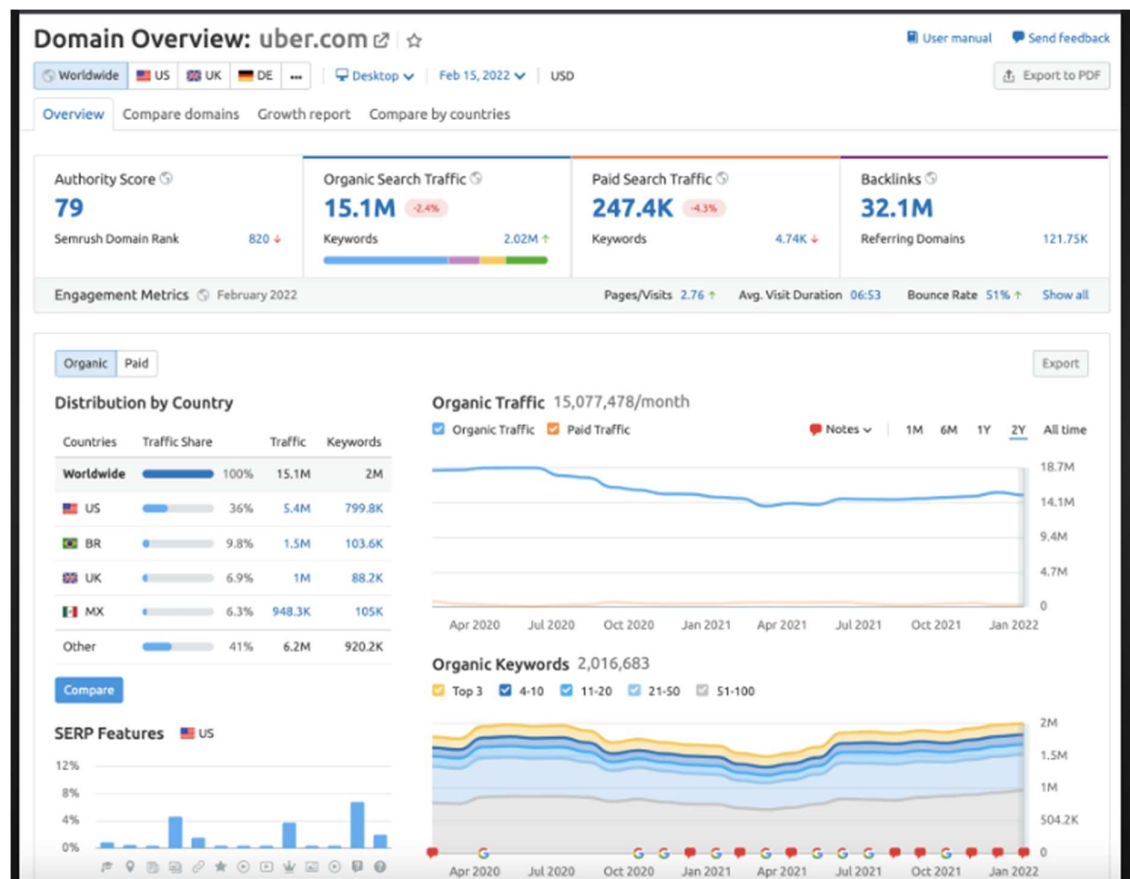
	Page ?	Pageviews ?	Unique Pageviews ?	Avg. Time on Page ?	Entrances ?	Bounce Rate ?	% Exit ?	Page Value ?
		5,483 % of Total: 100.00% (5,483)	4,520 % of Total: 100.00% (4,520)	00:47:55 Avg for View: 00:47:55 (0.00%)	3,412 % of Total: 100.00% (3,412)	<0.01% Avg for View: <0.01% (0.00%)	62.23% Avg for View: 62.23% (0.00%)	\$0.00 % of Total: 0.00% (\$0.00)
☐	1.	1,519 (27.70%)	1,321 (29.23%)	01:05:37	1,273 (37.31%)	1.41%	75.91%	\$0.00 (0.00%)
☐	2.	430 (7.84%)	351 (7.77%)	00:39:40	317 (9.29%)	1.85%	52.33%	\$0.00 (0.00%)
☐	3.	90 (1.64%)	77 (1.70%)	00:30:33	20 (0.59%)	0.00%	55.56%	\$0.00 (0.00%)
☐	4.	61 (1.11%)	53 (1.17%)	00:50:33	34 (1.00%)	2.70%	62.30%	\$0.00 (0.00%)
☐	5.	42 (0.77%)	9 (0.20%)	00:20:53	9 (0.26%)	0.00%	21.43%	\$0.00 (0.00%)
☐	6.	30 (0.55%)	27 (0.60%)	00:11:48	19 (0.56%)	0.00%	46.67%	\$0.00 (0.00%)
☐	7.	25 (0.46%)	10 (0.22%)	00:49:13	8 (0.23%)	0.00%	28.00%	\$0.00 (0.00%)
☐	8.	25 (0.46%)	15 (0.33%)	00:46:56	15 (0.44%)	0.00%	60.00%	\$0.00 (0.00%)
☐	9.	25 (0.46%)	8 (0.18%)	00:39:11	0 (0.00%)	0.00%	4.00%	\$0.00 (0.00%)
☐	10.	22 (0.40%)	6 (0.13%)	00:43:57	0 (0.00%)	0.00%	4.55%	\$0.00 (0.00%)

1. Content analytics tools

[Content marketing teams](#) are catching onto what media publishers have known for years: if you're publishing a lot of content, you'd better have a good measurement system in place. Given how difficult it can be for content-oriented people to wade through tools like Google Analytics, it's no surprise then that content analytics has emerged as a new category of web analytics tools.

These tools help content teams measure how various audiences consume and share their content. The resulting data lets them understand how to better engage their readers and stimulate conversions.

SEMRUSH



Parse.ly

Parse.ly is a [content marketing analytics platform](#) designed specifically for measuring the performance of content. It offers an alternative to general-purpose web analytics tools for content teams to find valuable insights that help them make decisions. Content teams can see real-time and historical data in a central dashboard, and they

can easily sort and filter it by type, author, channel, source, and more. Everyone on the content team, from directors to editors to writers, [can use it easily](#).

Parse.ly's approach to measuring content deviates from the generalized approach, which relies heavily on metrics like bounce rate. Instead, it [measures engaged time via a 'heartbeat'](#) to produce a metric called engaged minutes. This measures continuous engaged time even across videos and interactive content, which is conspicuously missing from most tools like Google Analytics.

Parse.ly also pulls in data from external channels like social media (Facebook, Twitter, LinkedIn, Pinterest, Reddit, Instagram), search engines (Google, Bing, Yahoo!, Yandex), and platforms like web, mobile, Apple News, Facebook Instant Articles, Google AMP, SmartNews, iOS, and Android apps to give a comprehensive picture of content performance.

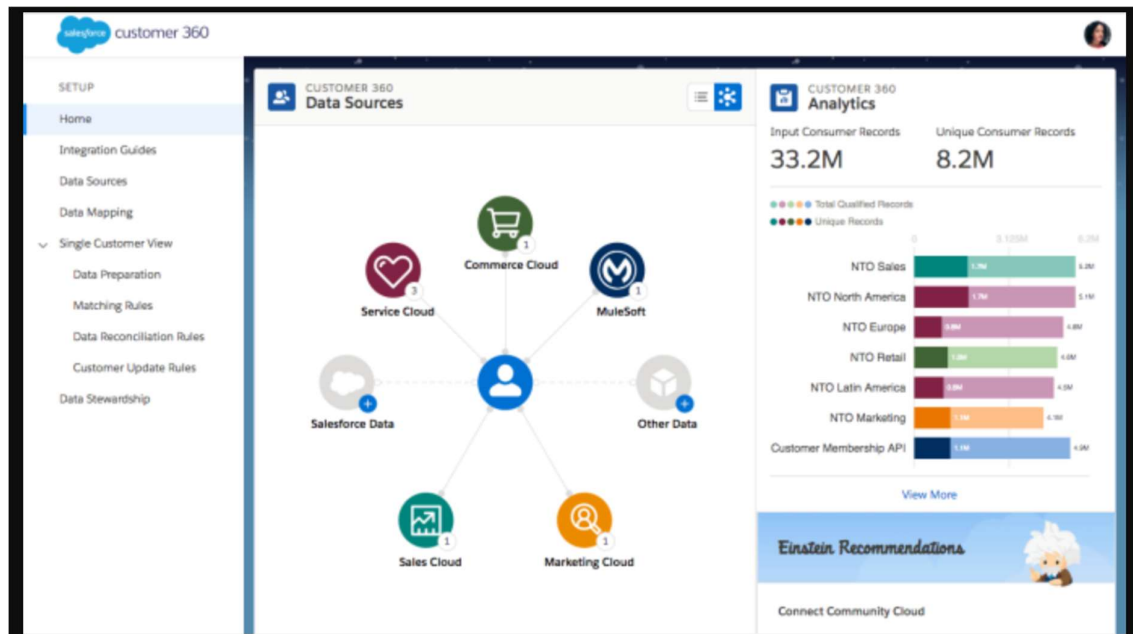


One of the add-ons available from Parse.ly [shows conversions](#) and attributions for actions, such as newsletter signups, subscriptions, registrations, lead captures, and ecommerce sales. See [more resources](#) and [how to get started](#) with content analytics.

2. Customer analytics tools

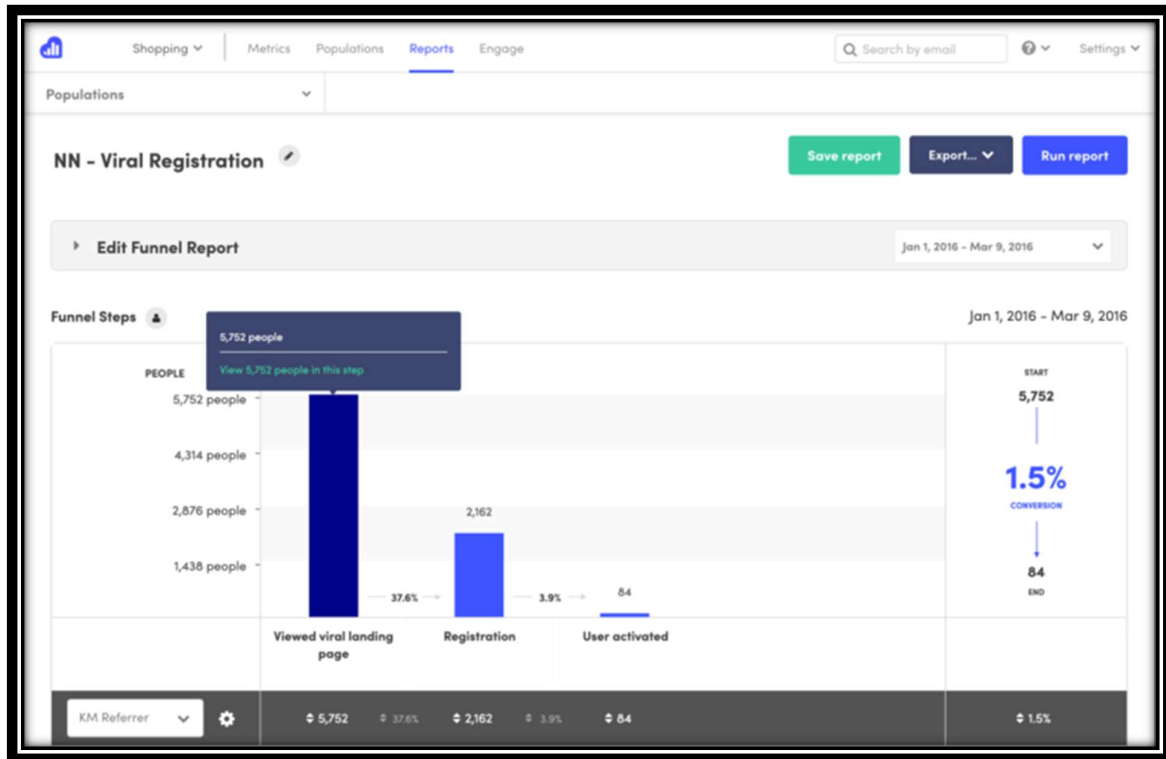
Customer analytics tools go deeper into customer behavior than other general web analytics tools. These tools pull in customer data from various mediums like web, mobile, email, and your product. You can create segments based on behavioral patterns, then predict and offer the products and services those distinct groups of customers might buy.

SALESFORCE



Kissmetrics

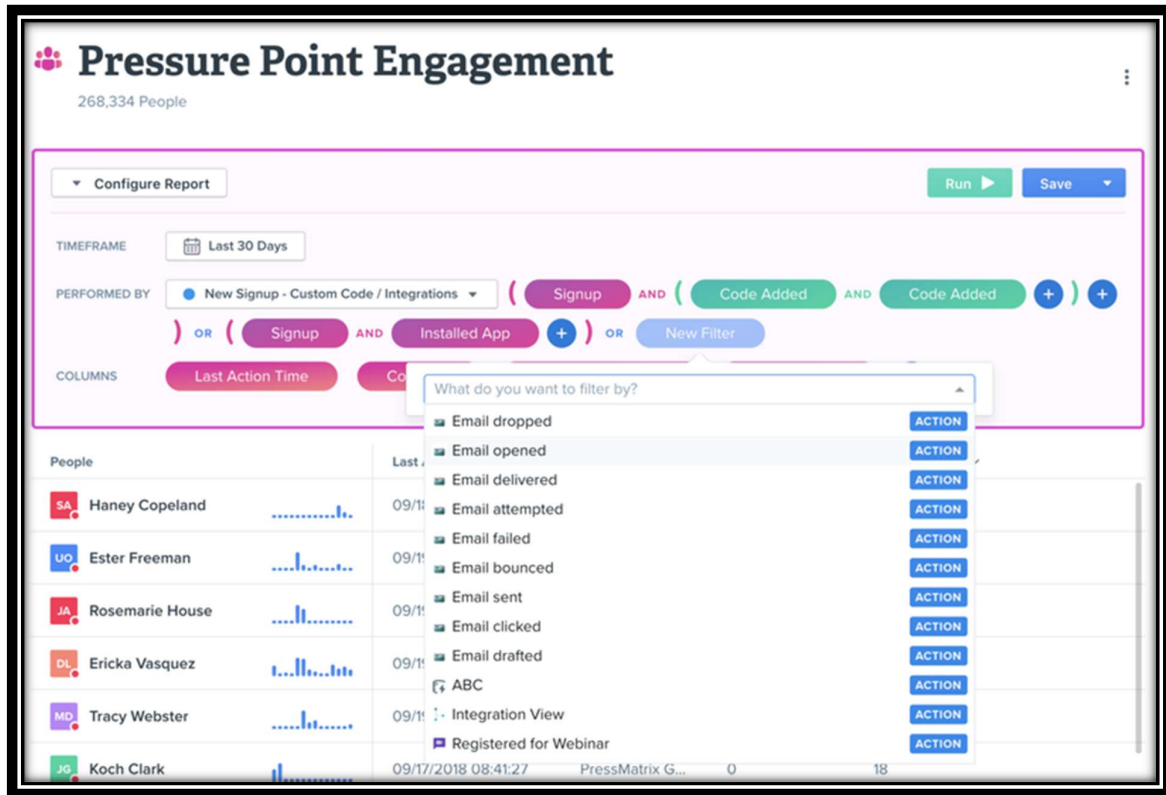
Kissmetrics excels in customer data for [ecommerce](#) and [SaaS companies](#). Both product and marketing teams can benefit from data on power users and other cohorts, insights into which features are most popular, and where your best customers come from. It helps identify groups of customers who spend more and make repeat purchases, as well as the products and offers they respond to.



Kissmetrics includes advanced business intelligence (BI) reporting with the ability to query the raw data directly with SQL. The company has helped over 10,000 companies with \$10 billion in transaction volume. Learn [how Kissmetrics differs from Google Analytics](#).

Woopra

Woopra is a less pricey alternative that shows what customers do between the time they hear of you, visit your site, and leave or take another action. It creates profiles for each individual visitor, taking into account the activity on your website, email, social, paid media, sales, technical support, help desk, and video, as well as engagement with your product. This is useful for marketing, sales, and product teams.

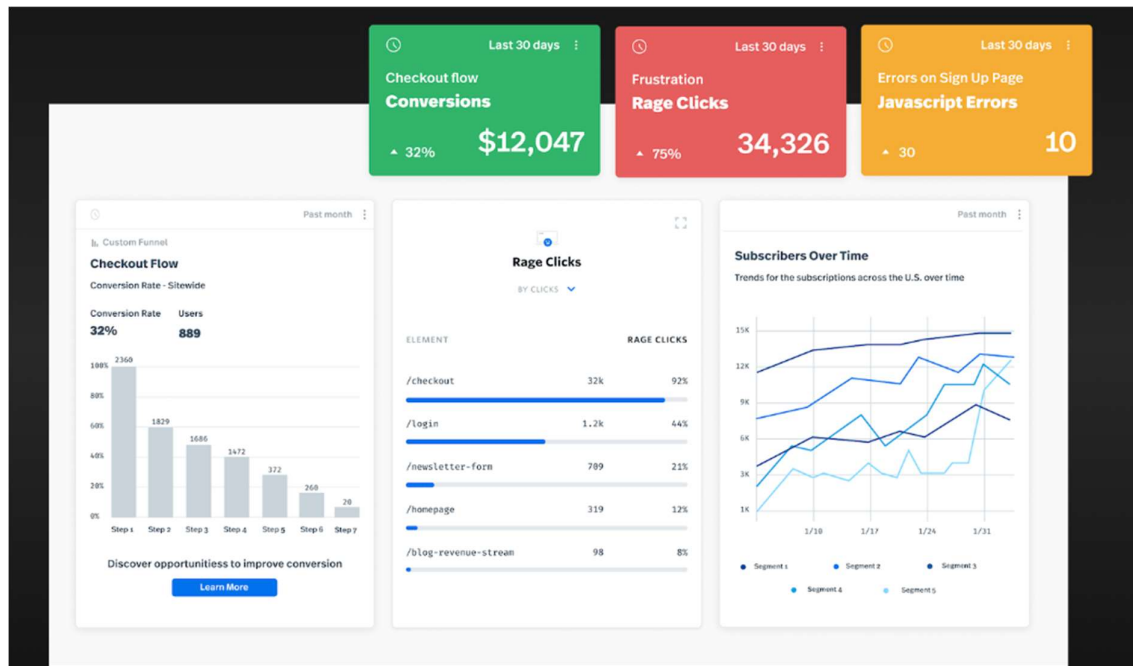


The tool also automatically deploys changes on your site based on triggers to help nudge customers along. Custom scripts let you display discount messages or newsletter opt-ins based on actions by the customer that you set as triggers. Learn more on the [Woopra blog](#) and see what it takes to [get started](#).

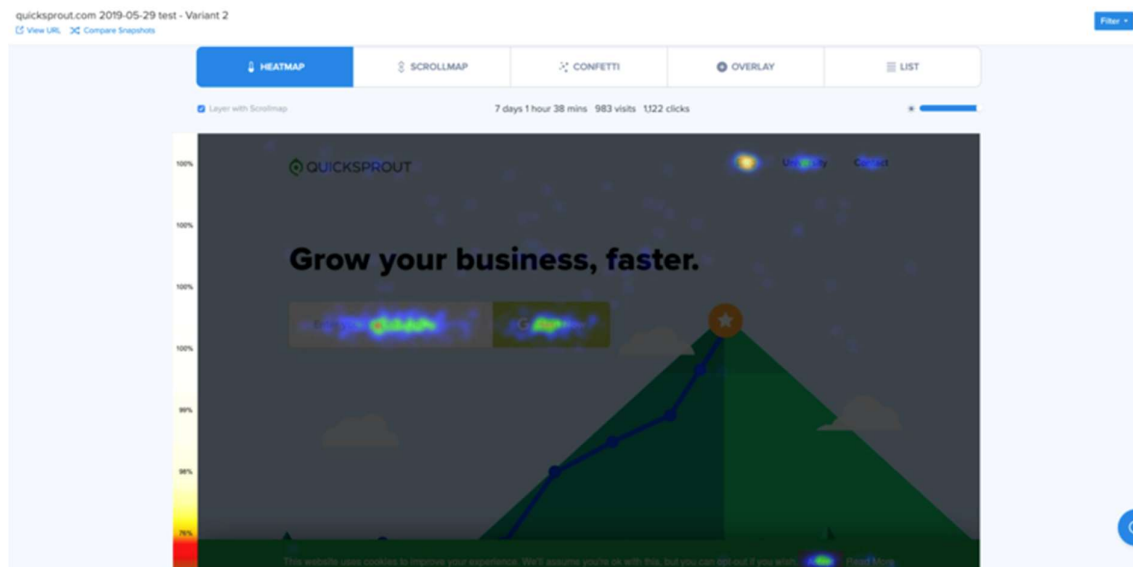
3. Usability (UX) analytics tools

Usability analytics tools are specialized tools that let businesses analyze what users do on a page. They record how people interact with the page and its elements, so an online marketing or product team can evaluate how different features are received. With these insights, businesses can then make changes to the page or user interface and see how those changes play out by measuring user behavior.

FullStory



Crazy Egg

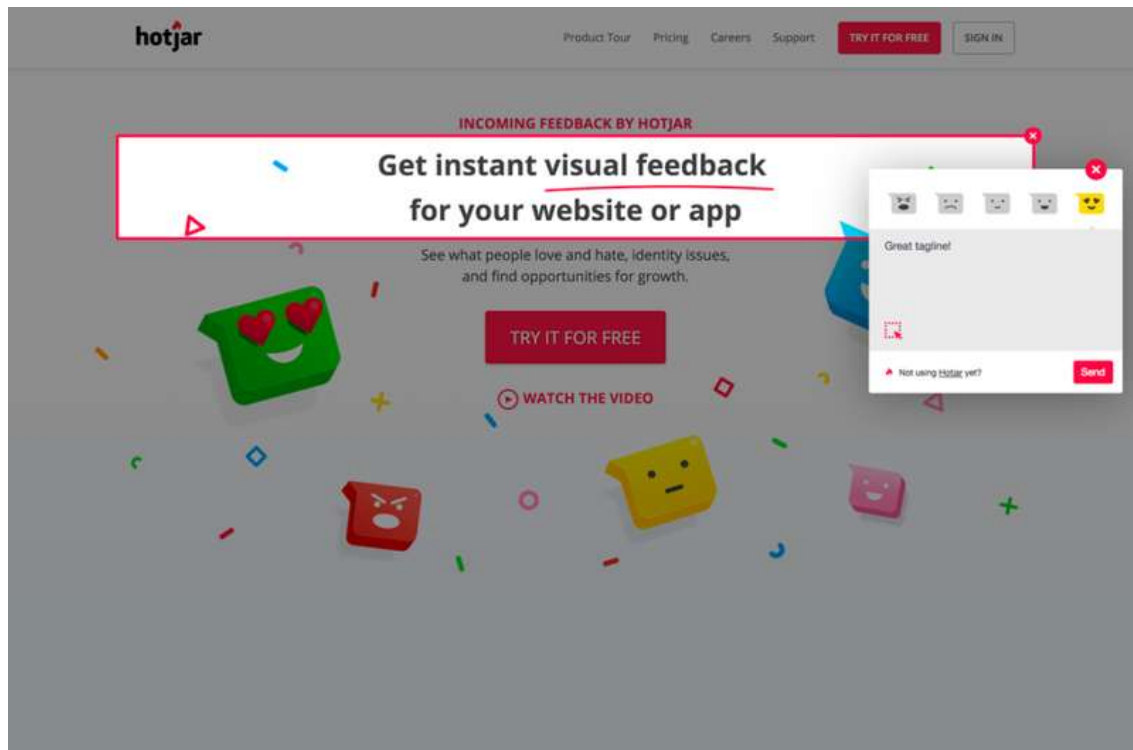


Crazy Egg focuses entirely on measuring how users navigate your website and landing pages. It's especially handy for optimizing conversion rates on pages where you want users to take action. It offers several technologies to help you understand how users are navigating your website, including heatmaps, scrollmaps, click reports, and user session recordings.

Crazy Egg also offers A/B testing, so you can compare heatmaps and data before and after changes on your web pages, for example. Check out the [Crazy Egg blog](#) and documentation for [getting started](#).

Hotjar

Hotjar gives you the option to collect actionable feedback from users in addition to data from heatmaps and visitor data recordings. You can use polls, surveys, and live feedback chat to query people on your website. This adds a qualitative dimension to the UX data you can collect, which is typically missing from other usability web analytics.



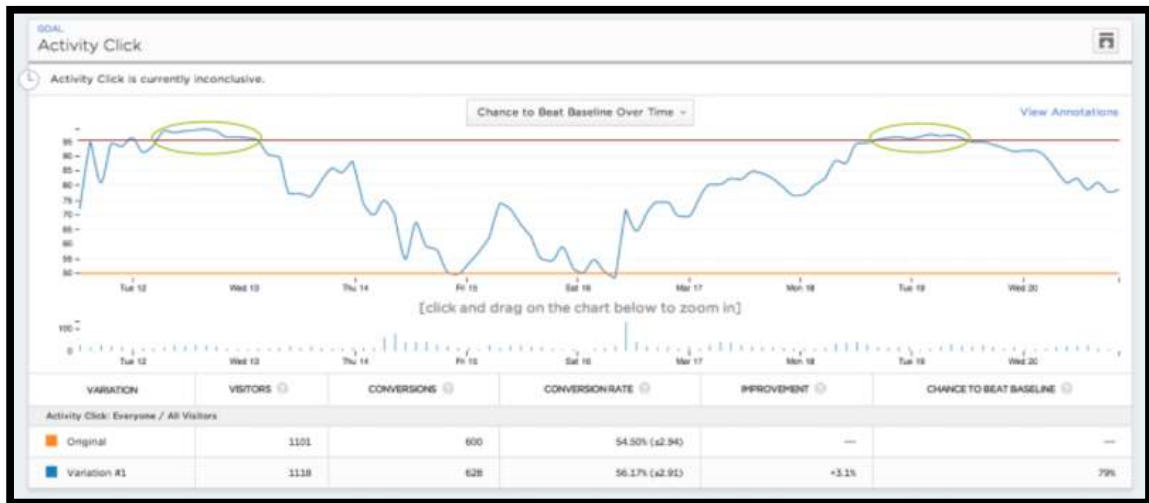
Hotjar does not offer A/B testing like Crazy Egg does, but you can [monitor pre-existing A/B tests](#). That means Hotjar could work for you if you already use a different tool that offers A/B testing. Check out its [resources page](#) and [what you need to get started](#).

4. A/B and multivariate testing tools

You can run the occasional simple A/B test using a general web analytics tool. But if you plan to test and optimize elements of your website regularly, you will benefit from a dedicated testing tool. And if you plan to run multivariate testing, which tests multiple variables on a page at once, you'll definitely need a tool like one of the following to accurately interpret results.

Optimizely

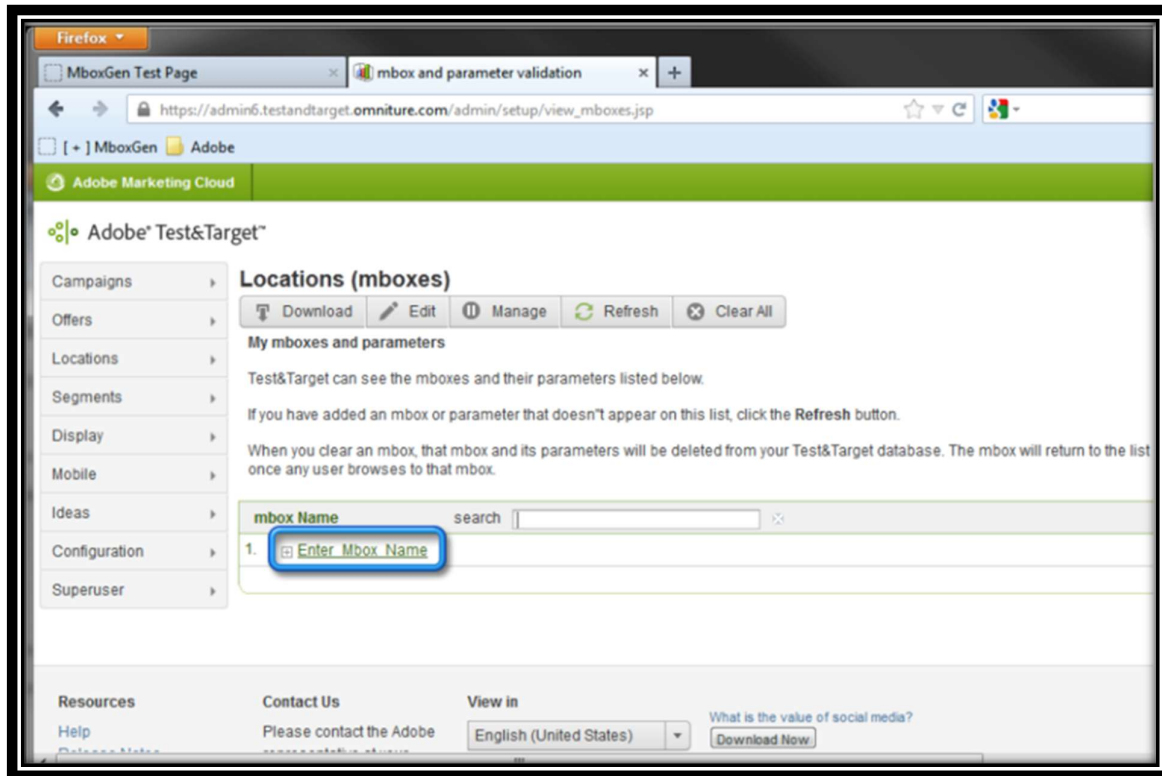
Optimizely offers testing analytics software to help businesses experiment and continuously improve their products and websites. It's used by engineers, product managers, marketers, and data scientists and also has a reputation for [catering to software developers](#) more than the other tools in this class. That said, [reviewers](#) say that it's simple to set up tests even with no technical know-how.



Optimizely notably includes a visual web editor, customer journey experimentation, and real-time analytics data. See Optimizely's [resource library](#) and [how to start using it](#).

Adobe Target

This tool is the way to go if you are already using the Adobe marketing stack and you want the best tool to integrate with Adobe Analytics. And it offers a very comprehensive list of capabilities. Adobe Target lets you build tests across other digital channels in addition to your website and app, including mobile and email.



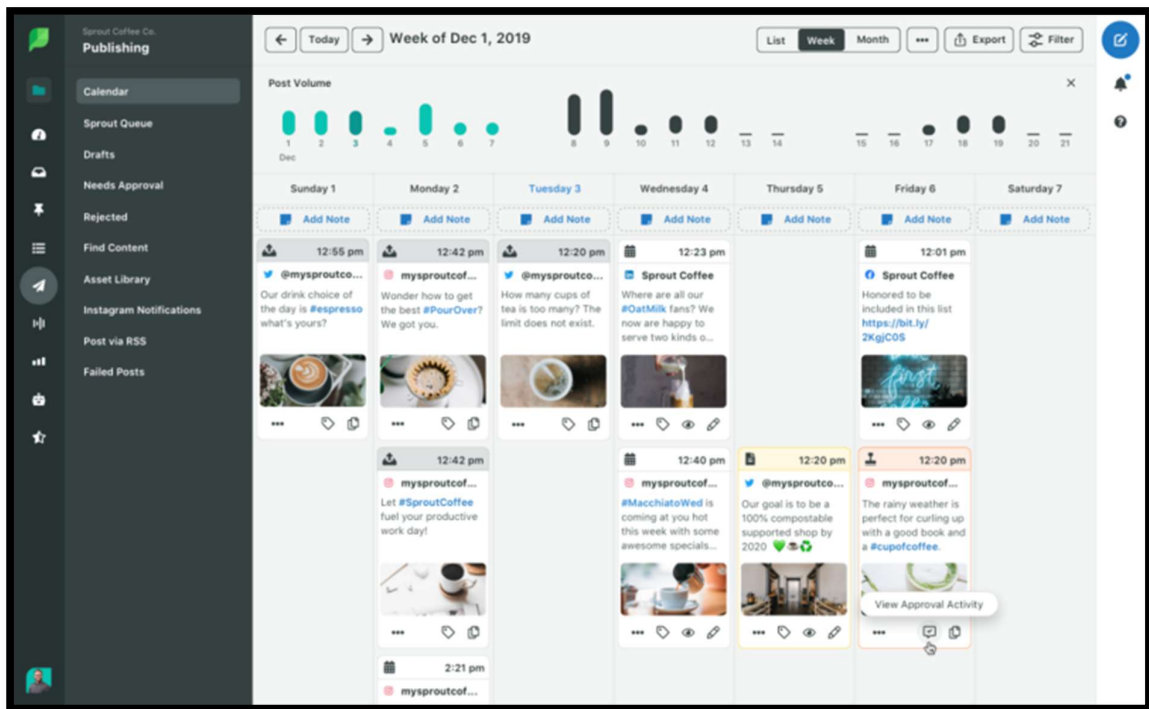
In fact, you can also set up personalized elements to deploy for users based on rules, profiles/segmentation, and behavioral targeting. See more on its [resources page](#) and learn about [getting started](#).

5. Social media analytics tools

Social web analytics tools are important for businesses focused on growing audiences and engagement on social media. While other tools may offer some features to help with this — for example, Parse.ly includes social media data for your content — you will need a dedicated tool if your business depends heavily on social media engagement. Some social analytics tools provide analysis only, while others let you manage your posts, too.

Sprout Social

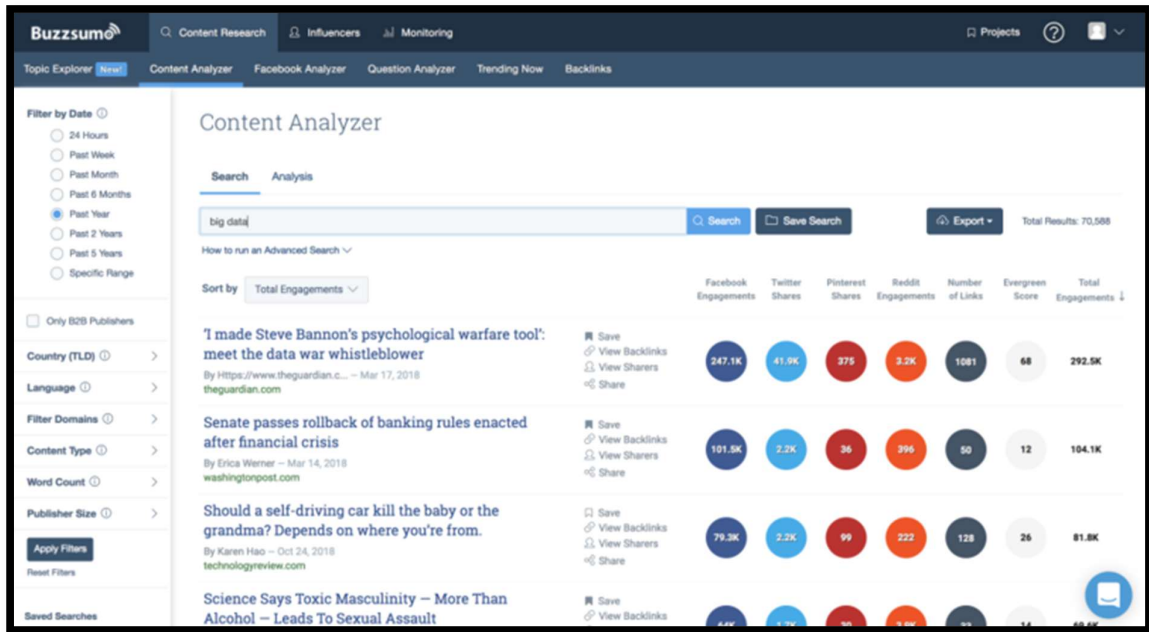
Sprout Social is an all-in-one social media management platform. You can use it to schedule and publish social media posts and monitor campaign performance. Keep your eye on keywords and hashtags, analyze your follower base, and even track paid marketing campaigns using the tool's key features.



Sprout Social lets you compare your performance on social media to that of your competitors. Its social media analytics surface insights on both the industry and the brand level, which you can apply to your product and business strategy. See more [resources](#) and how to [get started](#).

BuzzSumo

BuzzSumo is a social media research tool. It helps you analyze content to see what's gaining buzz in your industry. It shows what content readers respond to on social media, including what competitors are doing. BuzzSumo is useful to dig up topics in your industry, see whose content is most popular, and find targets for backlink and influencer outreach efforts.



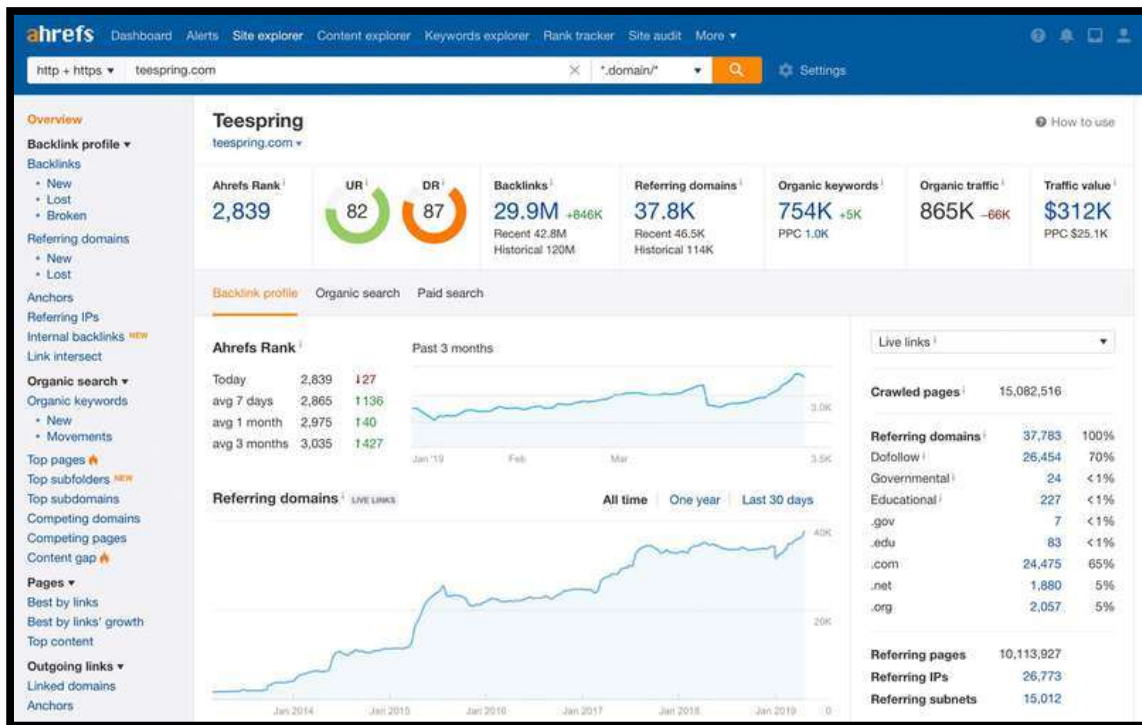
BuzzSumo also offers social media monitoring with alerts and an API to let you customize and integrate it into applications. Check out [the BuzzSumo blog](#) and documentation for [getting started](#).

6. SEO analytics tools

If you use search engine optimization to attract traffic from search engines, you most certainly need a specialized SEO tool in your web analytics tool stack. These tools provide data on keywords to point out new opportunities to gain traffic. They also give you data on backlinks pointing to a website. This is key because the number and quality of links pointing to a website are an important ranking factor for Google.

Ahrefs

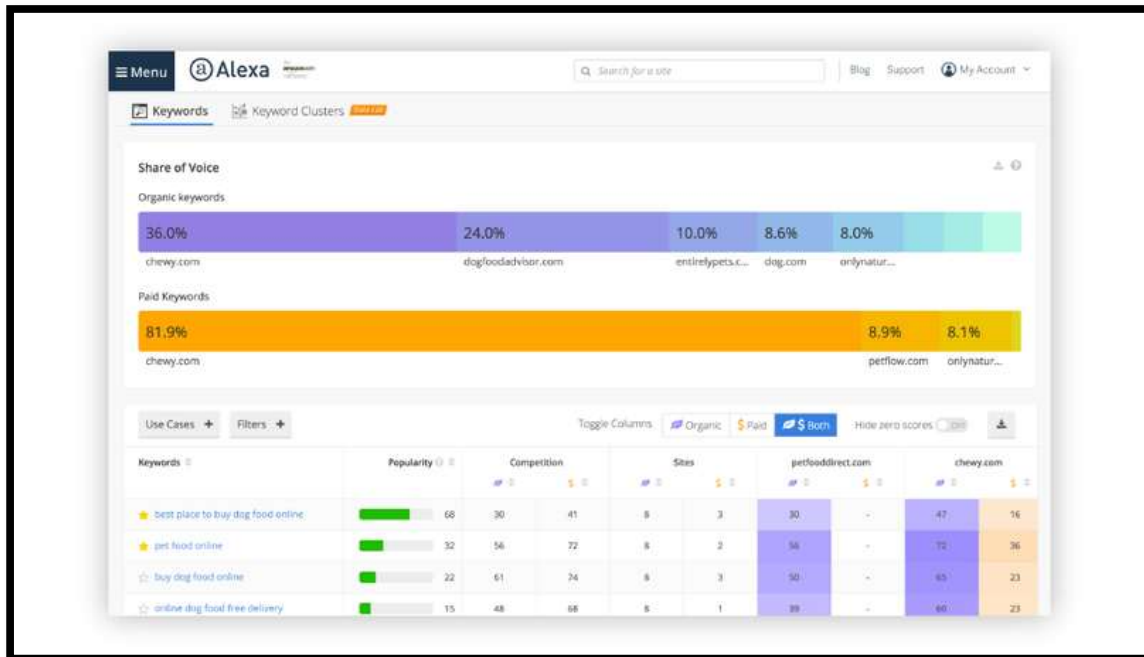
Ahrefs is a popular SEO analytics tool that offers both keyword research and backlink analysis, though [backlink analysis is Ahrefs' bread and butter feature](#). With the Ahrefs tool, you can see information about links pointing to any website. This includes data like the number of links over time, the quality of those links, and exactly which pages on a website link to yours. With this information, the SEO team can work to improve the website's backlink profile.



In addition to providing solid backlink data analysis, the tool is popular among SEO teams for keyword research. Learn more with Ahrefs' [resources page](#) and check out [popular questions](#) about the tool.

Alexa

Alexa, owned by Amazon and best known for its Alexa Rank metric, combines competitor intelligence with SEO and website analytics. Alexa's suite of tools includes backlink research, keyword research, site audits, and brand strength metrics. It also provides a unique feature where it shows you the likelihood that your site can rank for any given keyword, which is a great help when determining target keyword priorities.



Alexa particularly excels at competitor research and target audience analysis in the context of SEO. It gives a snapshot of related websites in an industry and then lets the user drill down to analyze websites by traffic sources, bounce rate, social engagement, backlinks, and keywords, among other factors. Check out [Alexa tutorials](#) and read more in the Alexa [blog](#).

7. General enterprise analytics tools

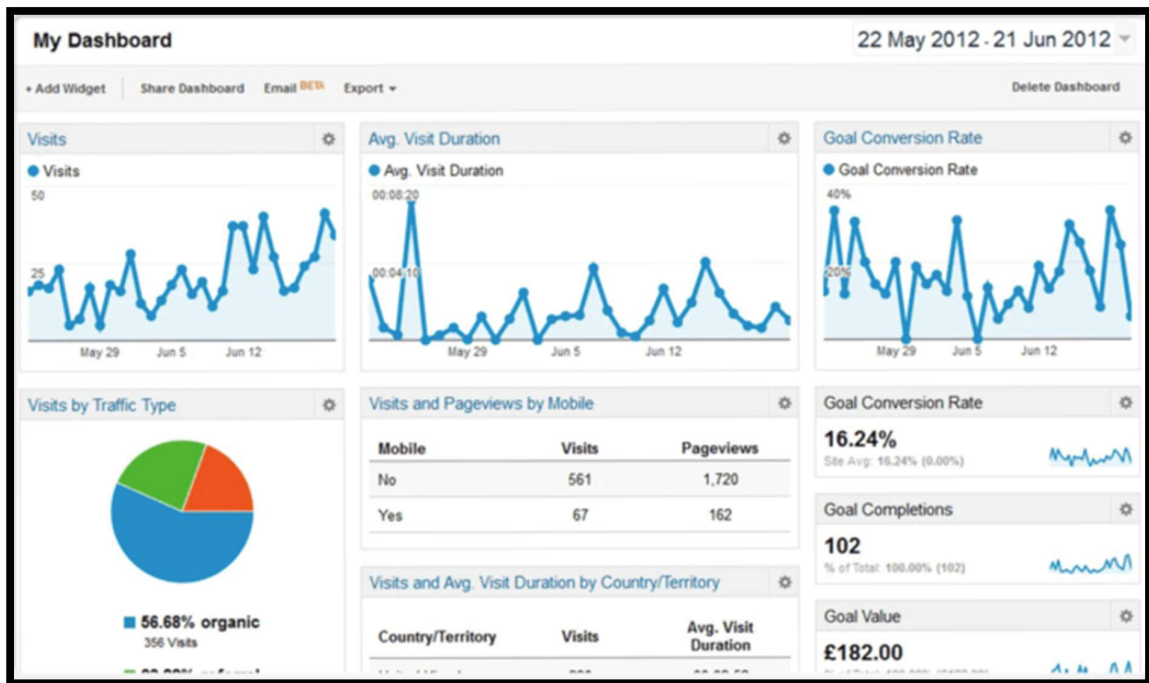
Those businesses that deal with large amounts of data face unique challenges and thus need tools designed to handle them. Enterprise-level web analytics tools are solutions designed especially for businesses with massive amounts of data — visitors that number in the hundreds of thousands or millions per month. At that point, you start to face challenges in separating the valuable data from the noise, data inaccuracies, and sampling errors and gaining the insights you need in a timely manner.

Parse.ly is an [enterprise-level content analytics](#) platform built for large content teams. But depending on the other activities your teams engage in, you may want to supplement it with a general enterprise analytics tool like the following.

Google Analytics 360

Google Analytics 360 is the premium version of Google Analytics. It includes advanced features that are suited to websites that deal with large numbers of users. This includes unsampled analytics data, meaning that, unlike regular Google Analytics, reports are all based on actual data rather than samples of data for greater accuracy. This enterprise-level web analytics tool integrates with other Google platforms, like Google Ads, as well

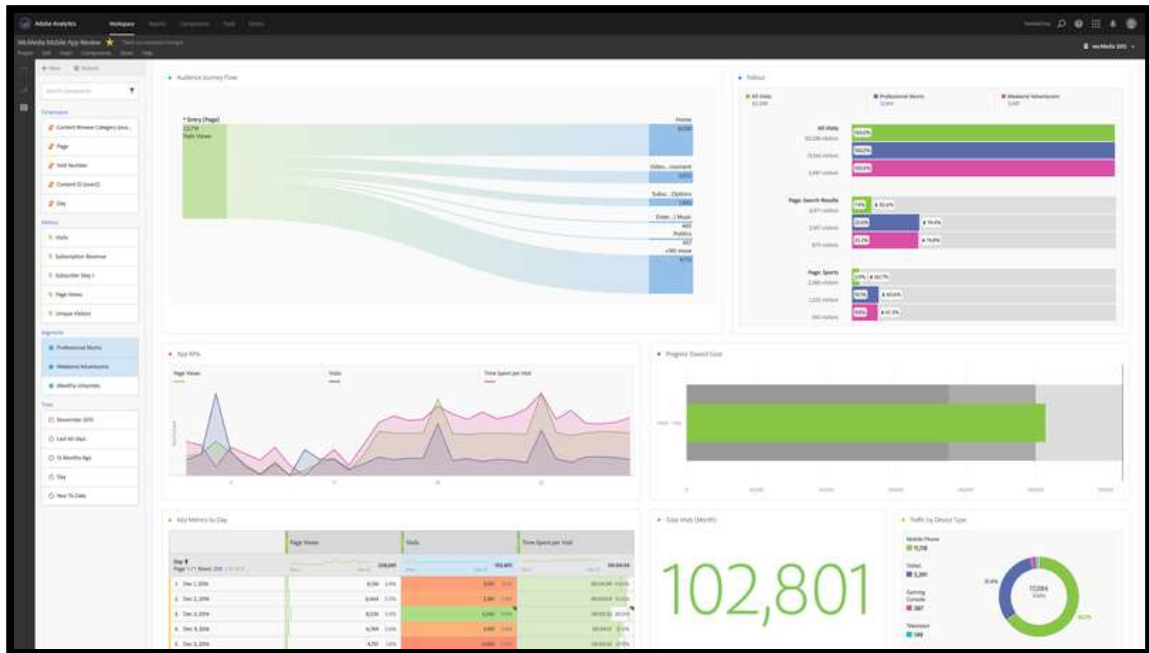
as other third-party tools, like the Salesforce Marketing Cloud. This [opens up opportunities](#) for campaign insights and remarketing.



Google Analytics 360 also offers advanced attribution modeling, which means a marketing team can use a more sophisticated way to measure the impact of various channels and touchpoints that lead up to a conversion. [Learn more about Google Analytics 360.](#)

Adobe Analytics

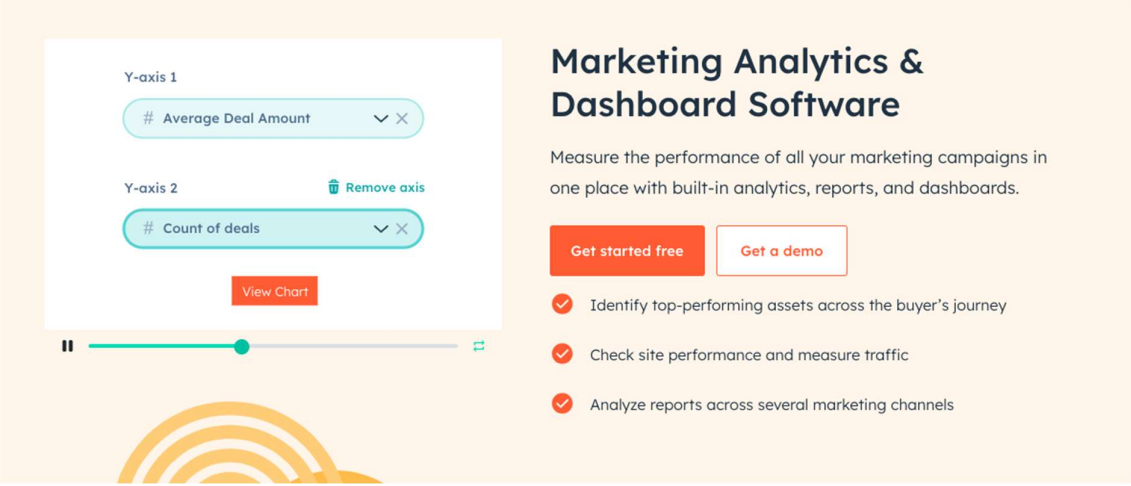
Adobe also offers an enterprise-level web analytics suite of tools. This solution is particularly useful for organizations that use Adobe Cloud and Adobe Target because it integrates easily with tools in the Adobe suite. Fans say that its drag-and-drop [reporting interface is easy to use](#). Similar to Google Analytics 360, Adobe Analytics can handle large volumes of data with no data sampling. But it also goes beyond the website-only focus of Google Analytics 360.



The platform also incorporates Adobe Sensei, which brings some benefits like a reduced time to discover statistically valid segments of customers. Data can be 'activated' into marketing workflows, meaning the platform can deploy audience segmentation, content optimization, and personalize the experience for the website visitor. The platform also allows broad data collection, even outside Adobe properties. Read more on [Adobe Analytics resources](#) and see what you need to [get started](#).

HubSpot

Everything that's measured is bound to grow. Measure and manage all your marketing-related data in one place and plan how it can translate to revenue with HubSpot's marketing analytics. Simply add your website's tracking code and let HubSpot do its magic. The [Marketing Analytics & Dashboard Software](#) collect essential data for you to make informed decisions. You can set automations based on the visitor's behaviour and trigger workflows that convert into sales or leads.

The image shows a software interface for 'Marketing Analytics & Dashboard Software'. On the left, there's a configuration panel with two Y-axis options: 'Y-axis 1' set to '# Average Deal Amount' and 'Y-axis 2' set to '# Count of deals'. There are dropdown arrows and a 'Remove axis' button for the second axis. Below this is a 'View Chart' button and a horizontal timeline slider. On the right, the title 'Marketing Analytics & Dashboard Software' is followed by a description: 'Measure the performance of all your marketing campaigns in one place with built-in analytics, reports, and dashboards.' Below this are two buttons: 'Get started free' and 'Get a demo'. A list of features follows, each with a red checkmark: 'Identify top-performing assets across the buyer's journey', 'Check site performance and measure traffic', and 'Analyze reports across several marketing channels'. The background is a light orange with a stylized sun graphic at the bottom.

If you rely on organic traffic to your website, you can measure the visitors with the tool. This helps you measure the quality of the visitors visiting your website. To improve the marketing campaigns, you can calculate the conversion rate based on the number of visitors' sessions. You can even identify key sources that get maximum conversions and focus only on those sources to generate leads or make more revenue.

You can get detailed marketing reports for all your operating channels. You can also view detailed reports related to all your social media campaigns, blogs and emails in one place. The visually appealing data in the dashboard is helpful and insightful for various teams to leverage. Lastly, as soon as you set up the marketing campaigns, the analytics dashboard starts collecting data for those campaigns. This is for those who don't have the technical expertise to set up the campaign from scratch.

8. Open source web analytics tools

Open source web analytics tools give you more freedom and control over your data than cloud-hosted solutions like those above. Your IT team can use and modify the source code to build your own customized in-house software, so you can store and analyze data how you prefer. This type of software is all on-premise because you will store it — and all the data in it — on your own servers. You are then responsible for maintaining and updating the software, too.

Among open source web analytics tools, two options stand out.

Matomo (formerly Piwik)

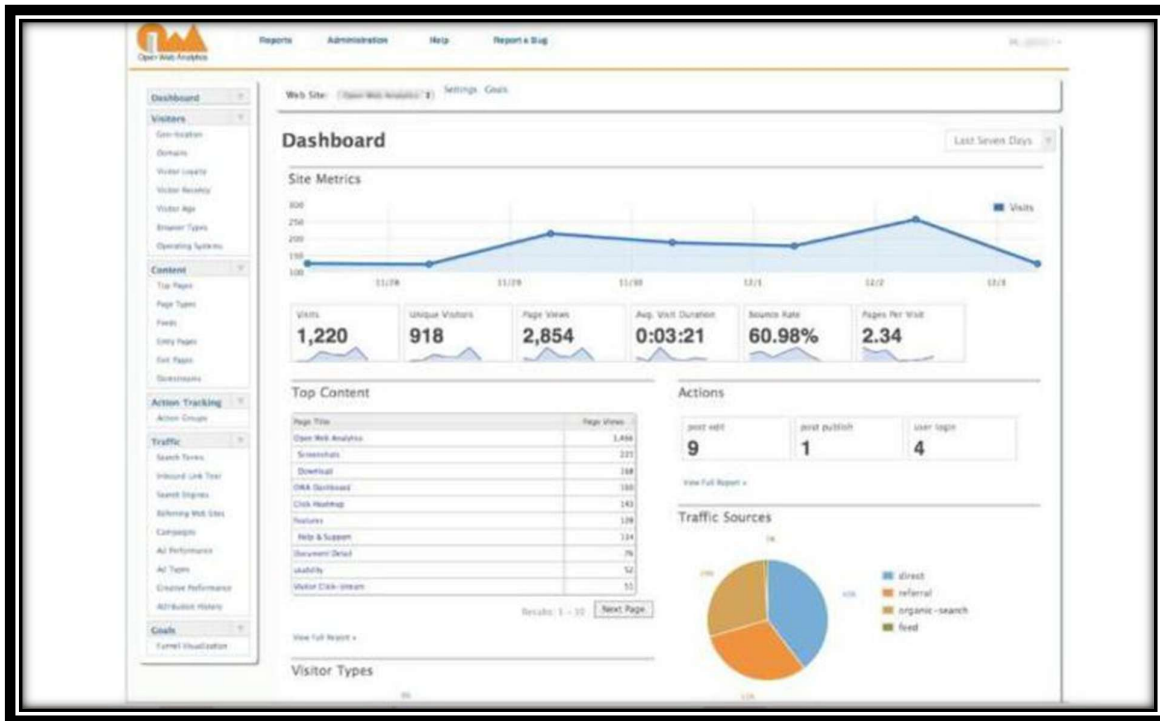
Matomo offers open source code with the freedom to modify it and share improvements among a vibrant community of developers. The software makes available a general analytics solution along with other more specialized features, like conversion funnels, visitor profiles, enhanced SEO, heatmaps, A/B testing, and intranet analytics.



Matomo also [doesn't use data sampling](#), which other tools like Google Analytics do. You can choose between cloud installation, which requires a subscription like the other analytics tools mentioned above, or [on-premise](#), which means you can install the software on your own server and make it your own. Either way, all [the data belongs to you](#).

Open Web Analytics

Close behind Matomo in terms of popularity, Open Web Analytics also offers many of the features you'd hope for in an open source all-in-one web analytics tool, plus a few more. Notably, this solution lets you see where people click on a page and how often people visit to measure loyalty. You can track an unlimited number of custom site actions.



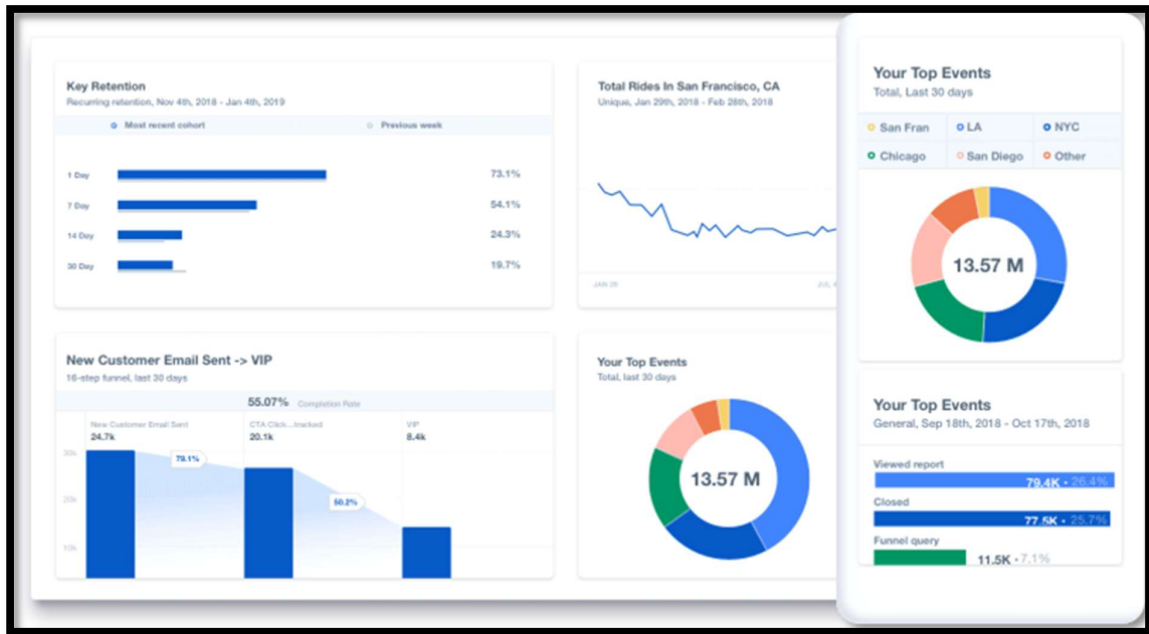
The software hooks you up with dozens of standard metrics, dimensions, and reports. Even though it's a free tool, remember, you'll need IT resources to set it up and keep it working. [View demos here.](#)

9. Product analytics tools

Product analytics tools measure how people interact with and use a digital technology product or app rather than a website. These analytics tools are used by product development teams to improve the user experience. They can see if users are interacting with a new feature, for example, or if it's causing friction for them, and then make changes accordingly. Some product analytics tools also measure data from your website in addition to that of your product.

Mixpanel

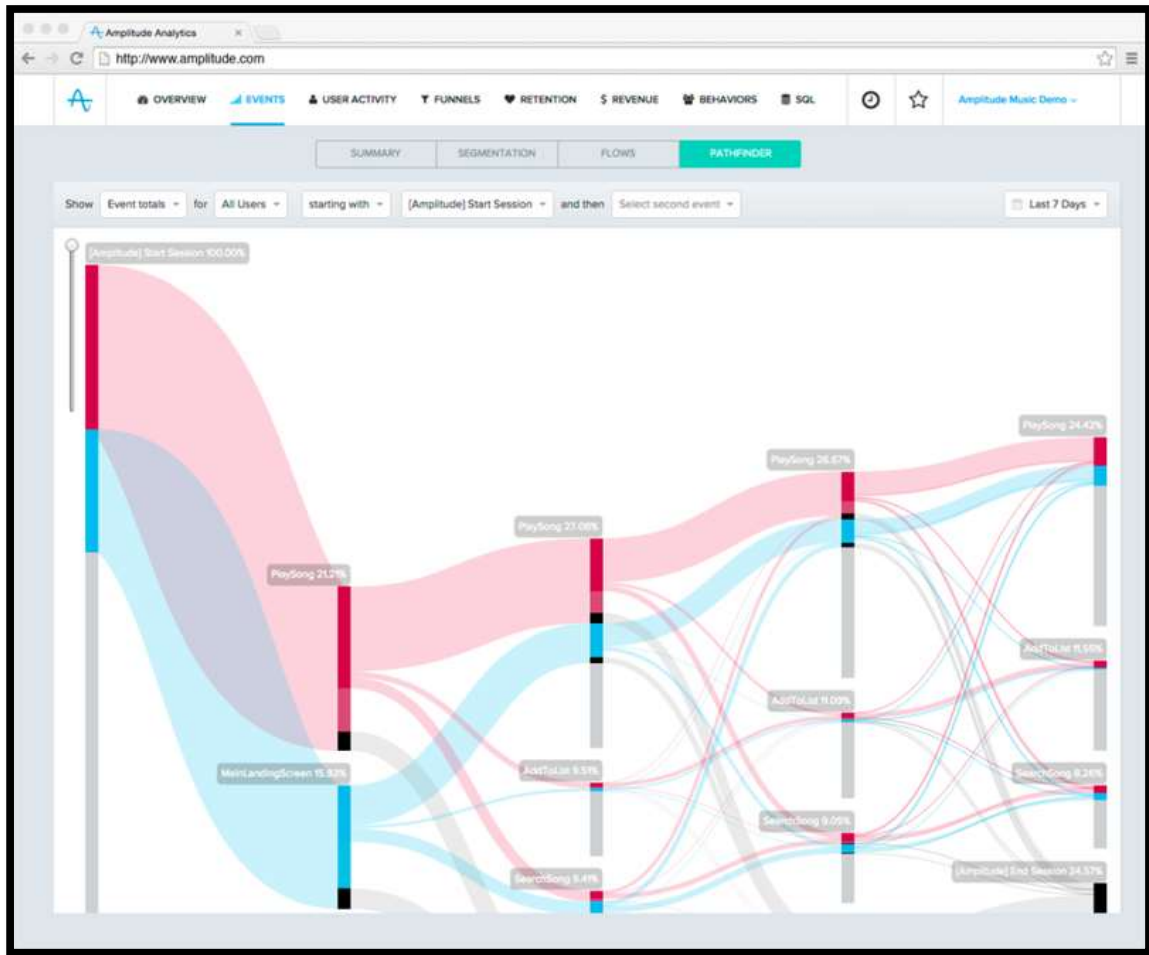
Mixpanel offers product analytics tools in addition to more general web analytics tools. Its [product-focused tools](#) pull out data on how customers are using a product in real-time, highlighting trends over time. Then, automated analysis tools help product managers understand why those trends may be occurring. To continually improve the product, the team can set up cohorts of users and test how product changes resonate with those users.



According to information on Capterra, over 26,000 companies — including [30% of The Fortune 100](#) — use Mixpanel. Customers say that they appreciate being able to run tests without involving developers. See [resources](#) and [how to get started](#).

Amplitude

Like Mixpanel, Amplitude offers self-serve product analysis, so you use a [data-driven product strategy](#) without technical know-how. However, unlike Mixpanel, Amplitude is entirely focused on product analytics. Its data collection and analysis extend through the entire user flow from onboarding through trial, purchase, use, and churn. It maps out all the possible steps customers take in the product to find points with high drop-offs. This points to changes you can make for a big impact.



One of the core beliefs of the company behind the tool revolves around finding and tracking a [North Star Metric](#) for your product. This means aligning the team and their efforts around one measurement that is key to product success.

The company says that [over 30,000 teams use Amplitude](#) to grow their digital businesses. To learn more about the tool, check out the Amplitude [resources library](#), [blog](#), and tools to [get started](#).

Which are the best web analytics tools for you?

As you can see, there are a lot of web analytics tools — and the information that each provides is very attractive. Who doesn't want to know as much as they can about their customers?

But adding more tools isn't always better. Tool fatigue can detract from their usefulness because each tool you add also adds a cost — in the form of time, money, and your attention.

So how can you know which web analytics tools are best for you?

The short answer is that it comes down to your team's priorities. If you are a small business still, you may only need one tool, like Google Analytics. If you are focused on growing via SEO, then, by all means, equip your team with a specialized SEO tool. And if you have reached scale with your content marketing and you want to improve your ROI, then it's time to consider a dedicated [content marketing analytics tool](#) like [Parse.ly](#).

[Talk to a content analytics expert](#) at Parse.ly today.

Setting Up Goals and Conversion Rate in Google Analytics 4 (GA4)

In **GA4**, the older Universal Analytics concept of **Goals** has largely been replaced by **Key Events**. A key event represents an important user action that indicates business success—for example, a purchase, registration, form submission, or app download.

1. What is a Goal in GA4?

A **goal** is an action that you want visitors to complete.

Examples:

Website objective	Goal / Key Event
E-commerce	Purchase
Generate leads	Form submission
Get registrations	Sign-up
Get enquiries	Contact form
Increase engagement	Video watched
Get downloads	PDF/software download
Increase calls	Phone-number click

Example:

Suppose an e-commerce website gets 1,000 visitors and 50 visitors purchase a product.

Conversion rate = $50 \div 1,000 \times 100 = 5\%$

$37\% = \frac{37}{100} = 0.37$

Percent of the whole

%

Percent of the whole

Give feedback

2. Creating a Goal/Key Event in GA4

The basic process is:

User action → Event → Mark as Key Event → Analyze conversion rate

Step 1: Identify the important action

First decide what action represents success.

For example:

A customer completes the checkout and reaches the purchase confirmation page.

GA4 may already collect an event such as **purchase** for a properly implemented e-commerce website.

Step 2: Go to Events

In GA4:

Admin → Data collection and modification → Events

Here you can see the events being collected by your website/app.

Examples:

- page_view
 - session_start
 - scroll
 - click
 - sign_up
 - generate_lead
 - purchase
-

Step 3: Mark the event as a Key Event

If an event represents an important business conversion:

Admin → Data display → Key events

You can mark the relevant event as a **Key Event**.

For example:

purchase → Key Event

This tells GA4:

"A purchase is an important action for this website."

3. Example: Creating a Lead Goal

Imagine a college website has a "**Contact Us**" form.

A visitor fills out the form.

The website sends this event:

generate_lead

You can mark:

generate_lead → Key Event

Now GA4 can report how many users generated leads.

Example data

Metric	Result
Users	10,000
Users who submitted form	500
Key Event	generate_lead
Conversion rate	5%

4. Conversion Rate in GA4

Conversion rate tells us **what percentage of users or sessions resulted in a desired action**.

A simple user conversion-rate calculation is:

Conversion Rate = Converting Users ÷ Total Users × 100

Example

Suppose:

- Total users = 20,000
- Users who purchased = 800

Conversion rate:

$$800 \div 20,000 \times 100 = 4\%$$

So the website has a **4% user conversion rate**.

Note: GA4 can report different conversion-rate metrics depending on the report/exploration and whether you are looking at **user** or **session** behavior, so students should always check which denominator the particular metric uses.

5. Goal/Key Event Reports in GA4

After setting up key events, you can analyze them through GA4 reports and Explorations.

Important reports/areas

Reports → Engagement → Key events

You can examine:

- Number of key events
- Event count
- Users
- Revenue associated with events
- Event rate/conversion-related metrics

You can also use:

Explore → Free-form exploration

to perform deeper analysis.

6. Example of a Goal Report

Suppose an online shoe store has three important actions:

1. Product enquiry
2. Add to cart
3. Purchase

GA4 data might look like:

Key Event	Users	Key Events
-----------	-------	------------

Generate lead	1,000	1,200
---------------	-------	-------

Add to cart	3,000	4,500
-------------	-------	-------

Purchase	500	600
----------	-----	-----

The marketing team can determine which campaigns and channels are generating valuable actions.

7. Conversion Funnel Example

Consider an e-commerce website:

1,00,000 visitors

↓

30,000 viewed products

↓

10,000 added products to cart

↓

4,000 started checkout

↓

2,000 completed purchase

The marketer can identify where users are dropping out.

For example:

Product View → Add to Cart → Checkout → Purchase

If many users add products to the cart but don't purchase, the company can investigate:

- High shipping charges
- Complicated checkout
- Payment problems
- Lack of trust
- Unexpected costs

This is one of the major uses of GA4 conversion analysis.

8. Goal Reports vs GA4 Key Event Reports

Universal Analytics GA4

Goals	Key Events
Goal completions	Key event count
Goal conversion rate	User/session conversion-related rate metrics
Goal URL	Event-based measurement
Destination goal	Event triggered when desired action occurs
Funnel reports	Explorations / funnel analysis
Goal value	Event value / revenue

Important point for students

Universal Analytics used "Goals" as a major conversion concept. GA4 uses an event-based model, with important events designated as "Key Events."

9. Real-life Digital Marketing Example

Suppose **Nike's website** runs a Google Ads campaign.

The campaign receives:

50,000 users

Of these:

- 5,000 view products
- 2,000 add products to cart
- 1,000 begin checkout
- 500 purchase

The marketer can mark **purchase** as a Key Event.

Then the marketer can investigate:

Google Ads → Website → Product → Cart → Checkout → Purchase

This allows the marketer to evaluate whether advertising traffic is actually producing business results—not merely website visits.

10. Simple Exam Definition

Goal/Key Event:

A goal in digital analytics represents a desired action performed by a website or app visitor, such as purchasing a product, submitting a form, registering, or downloading a file. In GA4, important events are designated as **Key Events**.

Conversion Rate:

Conversion rate is the percentage of users or sessions that complete a desired action.

Goal/Key Event Report:

A GA4 report used to analyze important user actions, their frequency, users performing them, and their relationship with marketing channels and website performance.

Easy flow to remember

Set up Event → Mark as Key Event → Collect Data → Calculate Conversion Rate → Analyze Report → Improve Marketing

A/B and Multivariate Testing in Digital Marketing

A/B and multivariate testing tools help digital marketers **optimize website performance, ad campaigns, and user experiences** by running experiments to determine what variations drive better results. Here's a breakdown of both:

A/B Testing (Split Testing)

- ✓ Compares two or more versions of a webpage, ad, or email (A vs. B).
- ✓ Measures which version performs better based on key metrics (CTR, conversions, etc.).
- ✓ Ideal for testing major changes like headlines, call-to-action buttons, or layouts.

Multivariate Testing (MVT)

- ✓ Tests multiple elements (e.g., headlines, images, colors) simultaneously.
- ✓ Identifies which combination of elements works best.
- ✓ Requires significant traffic to get statistically meaningful results.
- ✓ Useful for optimizing multiple aspects of a page at once.

Scenario: Improving Landing Page Conversions for an E-commerce Store

Imagine you run an **online store selling smart home devices**. You notice that your landing page for a new **smart thermostat** has a **high bounce rate** and **low conversion rate (5%)**. You decide to run A/B and multivariate tests to improve performance.



smart thermostat

high bounce rate and low conversion rate (5%)



PAGE 2

1 A/B Testing Example

Goal: Increase purchases by testing different CTA (Call-to-Action) buttons.

Test Setup:

- Version A (Control): Button text says "Buy Now" (original version).
- Version B (Variant): Button text says "Get 20% Off – Buy Today!"

Results After Testing (Traffic: 10,000 visitors)

- Version A: Conversion rate = 5% (500 purchases). ↑
- Version B: Conversion rate = 7% (700 purchases). ↑

✓ **Winner:** *Version B increased conversions by 40%, so we implement it permanently.*

2 Multivariate Testing Example

Goal: Find the best combination of a **headline** and **CTA button** to maximize sales.

Elements Tested:

- Headline Options:

- A: "The Smartest Thermostat for Your Home"
- B: "Save Energy & Money with This Thermostat"

- CTA Button Text Options:

- X: "Buy Now"
- Y: "Get 20% Off – Buy Today!"

Test Setup (4 Variations):

- A-X ("The Smartest Thermostat for Your Home" + "Buy Now")
- A-Y ("The Smartest Thermostat for Your Home" + "Get 20% Off – Buy Today!")
- B-X ("Save Energy & Money with This Thermostat" + "Buy Now")
- B-Y ("Save Energy & Money with This Thermostat" + "Get 20% Off – Buy Today!")

Results After Testing (Traffic: 20,000 visitors)

Version	Headline	CTA	Conversion Rate
A-X	The Smartest Thermostat for Your Home	Buy Now	5% ✓
A-Y	The Smartest Thermostat for Your Home	Get 20% Off – Buy Today!	7%
B-X	Save Energy & Money with This Thermostat	Buy Now	6%
B-Y	Save Energy & Money with This Thermostat	Get 20% Off – Buy Today!	9%

✓ **Winner:** B-Y ("Save Energy & Money with This Thermostat" + "Get 20% Off – Buy Today!")

☑ **Impact:** Increased conversion rate from 5% → 9% (80% increase!)

Attribution Model| Single-Touch| First Touch| Last Touch| Multi-Touch| Linear|Time-Decay

Attribution modeling, also known as channel attribution modeling, is the process of assigning credit to various marketing channels or touchpoints that have contributed to a user's conversion or action.

Attribution models in digital marketing help businesses understand which touchpoints contribute to conversions, allowing them to optimize their marketing efforts.

Marketing attribution is an absolute must for identifying which touchpoints in the customer journey most heavily impact conversions.

Customer journeys are now multi-channel, multi-device, and often non-linear.

Single-Touch Attribution Models

These models assign 100% of the credit to **one touchpoint**.

First-Touch Attribution

Assigns full credit to the **first interaction** a user has with a brand.

Best for awareness campaigns but ignores later interactions.

Last-Touch Attribution

Gives full credit to the **final touchpoint** before conversion.

Useful for direct-response marketing but ignores earlier influences.

Multi-Touch Attribution Models

These models distribute credit across **multiple touchpoints**.

Linear Attribution

Gives **equal credit** to **all touchpoints** in the customer journey.

Simple but may not reflect the true impact of each step.

Time-Decay Attribution

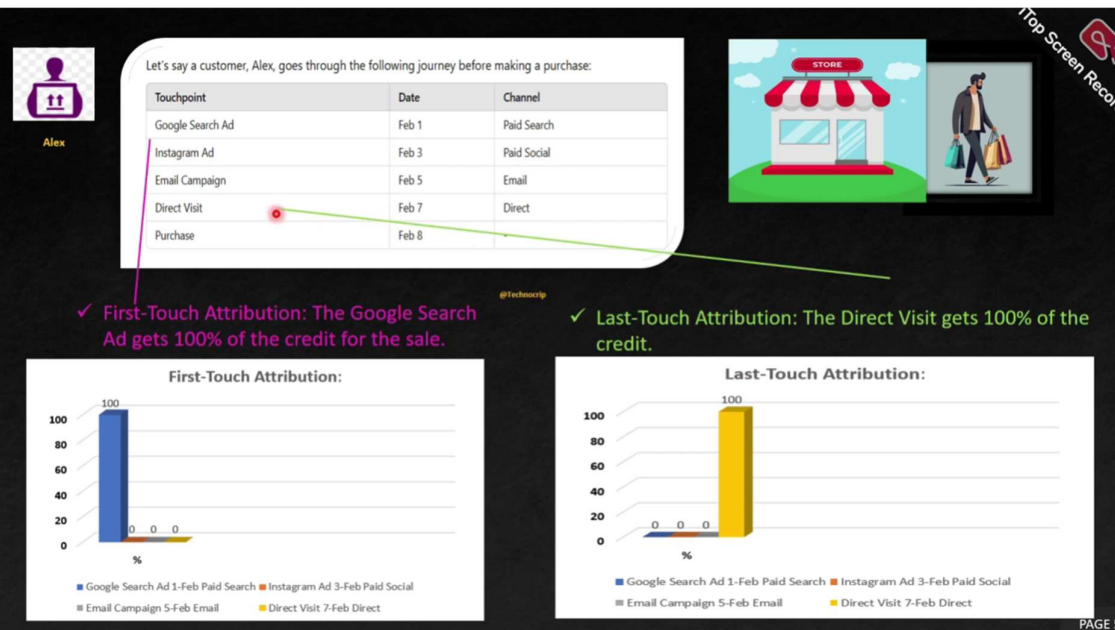
Assigns more credit to touchpoints **closer to conversion**.

Ideal for long sales cycles but undervalues initial interactions.

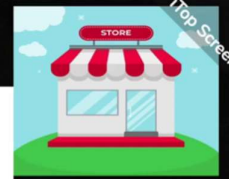
Position-Based (U-Shaped) Attribution

Typically assigns **40% credit** to the **first and last touchpoints** and splits the remaining **20% across middle interactions**.

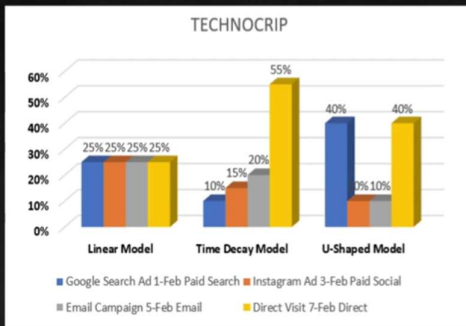
Good balance between awareness and closing interactions.



Using the same customer journey:



Touchpoint	Date	Channel	Linear Model	Time Decay Model	U-Shaped Model
Google Search Ad	Feb 1	Paid Search	25%	10%	40%
Instagram Ad	Feb 3	Paid Social	25%	15%	10%
Email Campaign	Feb 5	Email	25%	20%	10%
Direct Visit	Feb 7	Direct	25%	55%	40%



Linear Model: Each touchpoint gets 25% of the credit.

Time Decay Model: The Direct Visit (55%) gets the most weight as it's closest to the sale.

U-Shaped Model: The first (Google Search Ad - 40%) and last (Direct Visit - 40%) touchpoints receive the most credit.